



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

YJ40211 Legal, Social and Ethical Issues in
Information Systems (SBU's ISE 312)
Course Information

Ch 4: Intellectual Property

4.1 Principles, Laws, and Cases

4.2 Responses to Copyright

Infringement

4.2.1 Defensive and Aggressive Responses From the Content Industries

4.2.2 The Digital Millennium Copyright Act: Anticircumvention

4.2.3 The Digital Millennium Copyright Act: Safe Harbor

4.3 Search Engines and Online Libraries

4.4 Free Software

4.5 Patents for Inventions in Software



Principles, Laws, and Cases



Significant Cases

- File sharing: MGM v. Grokster
 - Grokster, Gnutella, Morpheus, Kazaa, and others provided peer-to-peer (P2P) file sharing services
 - The companies did not provide a central service or lists of songs
 - P2P file transfer programs have legitimate uses
 - Lower Courts ruled that P2P does have legitimate uses
 - Supreme Court ruled that intellectual property owners could sue the companies for encouraging copyright infringement
 - Businesses that encourage or provide tools for copyright infringement cannot operate legally in the United States

Principles, Laws, and Cases

Discussion Question

- *What do you think the impact would be on creative industries, such as music, movies and fiction novels, if copyright laws did not protect intellectual property?*

Principles, Laws, and Cases

Significant Cases

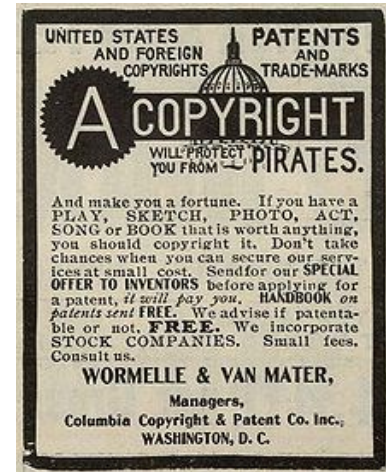
- “Look and feel”
 - Refers to features such as pull-down menus, windows, icons, and finger movements and specific ways they are used to select or initiate actions.
 - Reflects major creative effort by programmers.
 - In the 1980s and 1990s, some companies won copyright infringement suits against others whose software had similar look and feel
 - An appeals court ruled that menu commands are “a method of operation” excluded from copyright protection

4.2 Responses to Copyright Infringement

- Responses from the Content Industries
 - Expiration dates within the software: the software destroyed itself after that date
 - Dongles (a device that must be plugged into a computer port)
 - Copy protection that prevents copying
 - Activation or registration codes
 - Court orders to shut down Internet bulletin boards and Web sites
- Courts handed out severe penalties for organized, large-scale piracy
 - the owner of iBackup received a prison sentence of more than seven years and was ordered to pay restitution 补偿 of more than \$5 million after pleading guilty to illegally copying and selling software
 - a man who repeatedly recorded new movies on his camera in movie theaters and made pirate copies to sell received a sentence of seven years in jail

Responses to Copyright Infringement

- International Piracy
 - Some countries do not recognize or protect intellectual property
 - Piracy accounts for 42% of personal computer software in use worldwide (Business Software Alliance (BSA))
 - Countries that have high piracy rates often do not have a significant software industry
 - Is easier for a consumer to find a street vendor selling a U.S. movie on DVD, book or software than to find an authorized dealer
 - Many countries that have a high amount of piracy are exporting the pirated copies to countries with strict copyright laws



Responses to Copyright Infringement

- International Piracy
 - In China, personal computer manufacturers used to sell their machines without an operating system
 - people to bought cheap, unauthorized copies
 - In 2006, the Chinese government required that all PCs be sold with an authorized operating system preinstalled
 - The BSA gives a piracy rate of 20% for the United States

Responses to Copyright Infringement

- Banning, suing and taxing
 - Ban or delay copying technology via lawsuits
 - CD-recording devices (lawsuits by a group of companies including Disney)
 - DVD players
 - Portable MP3 players
 - The Recording Industry Association of America (RIAA) sued in 1998 and obtained a restraining order to stop Diamond Multimedia Systems from shipping its Rio machine, a portable device to play MP3 music files
 - Require that new technology include copyright protections
 - Tax digital media to compensate the industry for expected losses
 - taxes on personal computers, printers, scanners, blank DVDs, recorders, iPods, and cellphones

Responses to Copyright Infringement

- Digital Rights Management (DRM)
 - Collection of techniques that control uses of intellectual property in digital formats
 - The producer of a file has flexibility to specify what a user may do with it
 - Prevent saving, printing, making more than a specified number of copies, distributing a file, extracting excerpts, or fast-forwarding over commercials.
 - DRM enables the content seller to prevent lending, selling, renting, or giving away a purchased copy
 - Includes hardware and software schemes using encryption
 - Apple, Microsoft and Sony all use different schemes of DRM

Responses to Copyright Infringement

The Digital Millennium Copyright Act (DMCA)

- Congress passed the DMCA in 1998
- Two important parts:
 - Anticircumvention 防御欺骗
 - Prohibit circumventing (crack) DRM technological access controls and copy-prevention systems implemented by copyright owners in intellectual property
 - Safe harbor
 - Protect Web sites from lawsuits for copyright infringement by users of site
 - The site operators must make a good-faith attempt to keep infringing material off their sites

Responses to Copyright Infringement

- The DMCA vs. Fair Use, Freedom of Speech, and Innovation
 - Lawsuits have been filed to ban new technologies for **anticircumvention**
 - U.S. courts have banned technologies such as DeCSS even though it has legitimate uses, while courts in other countries have not
 - CSS: content scrambling system is a digital rights management (DRM) and encryption system employed on many commercially produced DVD-Video discs.
 - Protesters published the code as part of creative works (in haiku, songs, short movies, a computer game and art)
 - U.S. courts eventually allowed publishing of DeCSS, but prohibited manufacturers of DVD players from including it in their products

Responses to Copyright Infringement

- The Library of Congress decides on exemptions to the DMCA's anticircumvention provisions
- Smartphones, tablets, game machines, and other devices have mechanisms to prevent installation of software or use of services that the maker of the device does not supply or approve
 - Cracking such mechanisms is sometimes called *jailbreaking*, *unlocking*, or *rooting*
- The Library of Congress ruled in 2010 that it is legal to alter phones to install third-party software (e.g., apps) or to use an alternate service provider
 - the rule does not allow the same actions, for similar purposes, on other devices

Responses to Copyright Infringement

- **Safe Harbor**

- Industry issues "take down" notices per the DMCA
- As long as sites like YouTube and MySpace comply with take down notices they are not in violation
 - **Take down notices** may violate fair use, some have been issued against small portions of video being used for educational purposes
 - Copyright owners request removal of their content (and links to their content) by sending so-called takedown notices
 - It infers costs to the copyright owners to find all videos that infringe on their copyright
- Copyright owners argue that the sites should have the responsibility of filtering out copyright-infringing material

Responses to Copyright Infringement

- Safe Harbor

- Viacom sued YouTube and asked for \$1 billion in damages because it found 100,000 of its videos on YouTube
 - YouTube responded that it complied with the law: **it cannot always tell which are unauthorized**
- Universal Music vs. the video sharing Veoh (2011)
 - Veoh won the trial level and on appeal
 - However, Veoh declared bankruptcy; it cited the huge legal costs
- We have better technology now: the detection and removal of infringing material is now automated

Responses to Copyright Infringement

- Evolving Business Models
 - Organizations set up to collect and distribute royalty fees (e.g. the Copyright Clearance Center), users don't have to search out individual copyright holders
 - Apple iTunes provides legal means for obtaining inexpensive music and generate revenue for the industry and artists
 - Revenue sharing allows content-sharing sites to enable the posting of content and share their ad revenues with content owners in compensation

Responses to Copyright Infringement

- Evolving Business Models
- Cloud storage raises copyright issues: **Safe harbor in the cloud?**
 - Is copying legally purchased files to and from the cloud a fair use?
 - Will the companies operating the cloud services have any responsibility for unauthorized content their customers store and share?
 - Since copyright holders do not see what is stored, they do not have the option of sending takedown notices.

Responses to Copyright Infringement

Evolving Business Models

- What does not work
 - Zediva, a small startup in 2011, bought DVDs and rented the content (not the physical DVD) to customers legally. Court ordered Zediva to shut down.
 - Pirate Bay (Sweden, 2009): Four organizers of the Pirate Bay were convicted of contributory copyright infringement.
 - Megaupload:
 - operated from Hong Kong and New Zealand, with servers in several countries, including the Netherlands.
 - had 180 million registered users
 - claimed that it took down infringing material when notified to do so
 - The U.S. government shut Megaupload in 2012 (by legally seizing its domain names), and police in New Zealand arrested its founder and several employees

4.3 Search Engines and Online Libraries

- Search Engines
 - Copying is essential to many of the operations and services of search engines
 - Individuals and companies have sued Google for almost every search service it provides (Web text, news, books, images, and video)
 - Caching and displaying small excerpts is fair use
 - Creating and displaying thumbnail images is fair use
 - Google negotiated licensing agreements with news services to copy and display headlines, excerpts, and photos.
 - Trademarked search terms
 - Businesses pay search engine companies to display the business's ads when a user enters specific search terms
 - What if a business "buys" the name of another company?

Search Engines and Online Libraries

- Tools for authorized sharing
 - Many authors and artists are willing to share samples of their work on the Web
 - Creative Commons, a nonprofit organization, developed a spectrum of licensing agreements similar to the GNU General Public License for software
 - they provide a large degree of flexibility
 - Flickr is one of the largest users of Creative Commons licensing
 - Anyone who stores photos on Flickr can indicate what uses he or she permits



Search Engines and Online Libraries



- Books Online

- Project Gutenberg digitizes books in the public domain in the 1970s
 - Volunteers typed the entire text of the books
- Microsoft scanned millions of public domain books in University of California's library
- Google has scanned millions of books that are in the public domain and that are not; they display only excerpts from those still copyrighted
- Some court rulings favor search engines and information access; some favor content producers

4.4 Free Software



- What is free software?
 - *Free software* is an idea advocated and supported by a large, loose-knit group of computer programmers who allow people to copy, use, and modify their software
 - *Open source* - software distributed or made public in source code (readable and modifiable)
 - *Commercial software*, often called *proprietary software*, is normally sold in object code, the code run by the computer, but not intelligible to people.
 - The source code is kept secret.
 - Richard Stallman is the best-known founder and advocate of the free software movement.
 - Stallman began the GNU project in the 1970s

Free Software



- GNU project
 - Began with a UNIX-like operating system, a sophisticated text editor, and many compilers and utilities
 - Now has hundreds of programs freely available and thousands of software packages available as free software (with modifiable source code)
 - Advantages:
 - More people can use and benefit from a program
 - With source code available, any of thousands of programmers can find and fix bugs
 - Developed the concept of *copyleft*
 - A developer copyrights the program and releases it under a *copyleft* agreement that allows people to use, modify, and distribute it, or any program developed from it, but only if they apply the same agreement to the new work
 - GNU General Public License (GPL) implements copyleft

Free Software



- Linus Torvalds wrote the Linux kernel in 1991
- Major companies began to appreciate the benefits of open source
 - IBM, Oracle, HewlettPackard, and Silicon Graphics, used, supported, and marketed Linux
 - Shell and Home Depot adopted Linux
 - Dell sold PCs with Linux installed
- Other free software:
 - Firefox Web browser
 - Apache Web server
 - MySQL database server
 - Android OS (Linux based)



Free Software

- Disadvantages:
 - There is no technical support number to call for help
 - Because anyone can modify free software, there are many versions and few standards, creating a difficult and confusing environment for nontechnical consumers and businesses
 - Many businesses want to deal with a specific vendor from whom they can request enhancements and assistance
- Some of these weaknesses are fading now:
 - New businesses developed to support and enhance free software: Red Hat, Ubuntu, Oracle



redhat®

25

ubuntu®



Free Software

- Should all software be free?
 - Would there be sufficient incentives to produce the huge quantity of consumer software available now?
 - Would the current funding methods for free software be sufficient to support all software development?
 - How are free software developers paid?
 - Government grants to universities as a way of funding software
 - Concepts such as copyleft and the GNU Public License provide alternatives to proprietary software within today's current legal framework



Free as in Freedom

4.5 Patents for Inventions in Software

"Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title."

—U.S. Patent Law (Title 35 U.S. Code, Section 101)

- Patent law is extremely complex
- Patents protect inventions by giving the inventor a monopoly for a specified time period.
- Laws of nature and mathematical formulas cannot be patented
- Obvious inventions or methods cannot be patented
- Google, Apple, and Microsoft paid billions of dollars to buy thousands of wireless and smartphone patents

Patents for Inventions in Software

- A patent holder can build the patented device using the patented element.
- The patent holder may license others to do so for a license fee, or royalty.
- You register with the government. Can register in foreign countries.
US patent is issued by USPTO
 - Registration may take more than a year
- Patents generally last for 20 years
- Types - Utility, design, chemical, software, etc.

Patents for Inventions in Software

- **Manual of Patent Examining Procedure (MPEP)**

<https://www.uspto.gov/web/offices/pac/mpep/>

- Chapter 100 - [PDF Secrecy, Access, National Security, and Foreign Filing](#)
- Chapter 200 - [PDF Types, Cross-Noting, and Status of Application](#)
- Chapter 300 - [PDF Ownership and Assignment](#)
- Chapter 400 - [PDF Representative of Applicant or Owner](#)
- Chapter 500 - [PDF Receipt and Handling of Mail and Papers](#)
- Chapter 600 - [PDF Parts, Form, and Content of Application](#)
- Chapter 700 - [PDF Examination of Applications](#)
- Chapter 800 - [PDF Restriction in Applications Filed Under 35 U.S.C. 111; Double Patenting](#)
- Chapter 900 - [PDF Prior Art, Classification, and Search](#)
- Chapter 1000 - [PDF Matters Decided by Various U.S. Patent and Trademark Office Officials](#)
- Chapter 1100 - [PDF Statutory Invention Registration \(SIR\); Pre-Grant Publication \(PGPub\) and Preissuance Submissions](#)
- Chapter 1200 - [PDF Appeal](#)
- Chapter 1300 - [PDF Allowance and Issue](#)
- Chapter 1400 - [PDF Correction of Patents](#)
- Chapter 1500 - [PDF Design Patents](#)
- Chapter 1600 - [PDF Plant Patents](#)
- Chapter 1700 - [PDF Miscellaneous](#)
- Chapter 1800 - [PDF Patent Cooperation Treaty](#)
- Chapter 1900 - [PDF Protest](#)
- Chapter 2000 - [PDF Duty of Disclosure](#)
- Chapter 2100 - [PDF Patentability](#)
- Chapter 2200 - [PDF Citation of Prior Art and Ex Parte Reexamination of Patents](#)
- Chapter 2300 - [PDF Interference Proceedings](#)
- Chapter 2400 - [PDF Biotechnology](#)
- Chapter 2500 - [PDF Maintenance Fees](#)
- Chapter 2600 - [PDF Optional Inter Partes Reexamination](#)
- Chapter 2700 - [PDF Patent Terms and Extensions](#)
- Chapter 2800 - [PDF Supplemental Examination](#)
- Chapter 2900 - [PDF International Design Applications](#)

Patents for Inventions in Software

- Companies do not buy the patents because they need them for products they are developing
 - Companies buy patents so that they can sue other companies for patent infringement when the other companies sue them for patent infringement
 - A consortium including Apple, EMC, Ericsson, Microsoft, Research In Motion, and Sony paid \$4.5 billion for the Nortel 6,000 patents and patent applications encompassing technologies such as wireless, wireless 4G, data networking, optical, voice, Internet, and semiconductors

Patents for Inventions in Software

- In 1968, the Patent Office declared computer programs not patentable
- In 1981, the Supreme Court said that while software itself is not patentable because it is abstract, a machine or process that includes software, and in which the sole new aspect is the innovation implemented in the software, could be eligible for a patent
- In the following decades, the Patent Office issued thousands of patents, and the Federal Circuit court (which handles patent appeals) approved many
- Patents now cover encryption algorithms, data-compression algorithms, one-click shopping and other e-commerce techniques, copy-protection schemes, news feeds, location-based services for smartphones, delivery of email to cellphones

Patents for Inventions in Software

- The Patent Office has a backlog of more than 600,000 patent applications
 - not enough patent attorneys to review the patents and determine if a new software product would violate an existing patent
 - grants an estimated 40,000 software patents each year
- In *KSR v. Teleflex* (2007), the Supreme Court broadened the definition of “obvious” for rejecting patents
- In *Bilski v. Kappos* (2010), the Supreme Court reemphasized that a patent must not give control over an abstract idea or mathematical algorithm

Patents for Inventions in Software

- A few cases
 - Amazon.com generated a lot of criticism when it sued Barnesandnoble.com for violating its patent on one-click shopping
 - Paul Allen, co-founder of Microsoft,) sued several companies (Google, Facebook, Apple, eBay, Netflix, AOL, and others) for violating four early patents related to now widely used e-commerce and Web-viewing features.
 - A judge dismissed the suit in 2011.
 - The Patent Office reconsiders the patents
 - Apple won a patent case against a maker of Android phones for technology that allows a user to tap a touch screen to call a phone number that is in an email or text message
 - IBM sued Amazon for violating several of its patents on recommending books to customers based on their previous purchases
 - Amazon agreed to pay IBM a licensing fee

Patents for Inventions in Software

- Patent trolls
 - Some companies accumulate thousands of technology patents but do not make any products.
 - They license the patents to others and collect fees.
 - Intellectual Ventures has an estimated 30,000 patents and collected close to \$2 billion in license fees
 - If the patents themselves are legitimate (still an open question for many), this business model is not unreasonable
 - an inventor might have neither the skills for nor the desire to market a technology

Patents for Inventions in Software

- To patent or not?
- In favor of software patents
 - Reward inventors for their creative work
 - Encourage inventors to disclose their inventions so others can build upon them
 - Encourage innovation
 - Encourage the large investment often required to develop innovative systems and techniques

Patents for Inventions in Software

- To patent or not?
- Against software patents
 - In the current system, patents can stifle 使窒息 innovation, rather than encourage it.
 - The holder of the 1895 patent on an automobile sued Henry Ford
 - Many software developers come up with the same techniques independently, but patent law does not allow them to use their own invention if someone else has patented it
 - Cost of lawyers to research patents and risk of being sued discourage small companies from attempting to develop and market new innovations.
 - It is difficult to determine what is truly original and distinguish a patentable innovation from one that is not.